

Task 3 — Performance Profiling & Bottleneck Elimination

Submission-ready analysis plan based on the supplied Task 3 brief

Objective

Measure performance from the user's perspective and connect latency/errors with conversion and abandonment, then quantify the associated revenue impact and identify where speed is worth engineering effort.

Metric definitions

Conversion rate: converted sessions divided by eligible sessions.

Abandonment rate: abandoned sessions divided by eligible sessions.

Latency: user-observed request/step latency in milliseconds; report median and high percentiles such as p95/p99.

Error rate: error events divided by eligible requests/events.

Revenue impact: compare observed conversion/revenue against a clearly defined baseline or matched performance cohort. Do not claim causation from observational correlation alone.

Analysis approach

Segment results by latency bands, page/step, and error status. Reconcile event counts against the source system, check missing/duplicate records, and report uncertainty or limitations.

For each material finding, document the source field, calculation, and decision it triggers. Prioritize bottlenecks using both performance effect and financial impact.

Action framework

Investigate steps with high latency, high error rates, meaningful abandonment, and measurable revenue exposure. Validate proposed fixes with a controlled experiment or other defensible design before attributing causal impact.

Submission note

The accompanying CSV contains headers only because no real performance dataset was supplied with the screenshots. Replacing the template with the actual production extract is required for a defensible final submission.

Data file used

Task3_Performance_Profiling.csv is an analysis-ready CSV template. The task brief requires real production data and explicitly says not to use synthetic data, so no fabricated observations are included. Populate it with the actual source-system extract before claiming final measured results.